# Aditya Raut

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## Professional Summary

Software Engineer with 3+ years of experience building scalable distributed systems, and high-performance LLM infrastructure across Java, Python, and AWS. Proven track record modernizing enterprise architectures, processing 2M+ monthly financial transactions, deploying resilient multi-model AI integrations, and optimizing deep learning inference through custom KV-cache quantization and low-latency RAG pipelines.

## Technical Skills

**Languages:** Java, Python, Go, JavaScript, TypeScript, SQL, Bash/Shell, C/C++, HTML5/CSS3
**Backend & Frameworks:** Spring Boot, FastAPI, Django, Express.js, React, Node.js, SQLAlchemy, Pydantic
**Distributed Systems:** Microservices, Event-Driven Architecture, REST APIs, Apache Kafka, Kinesis Video Streams, Asynchronous Task Queues
**AI & LLM Engineering:** KV-Cache Quantization (INT4/FP16), PyTorch, CUDA, RAG, FAISS, MCP, Prompt Orchestration
**Databases & Caching:** PostgreSQL, MySQL, MongoDB, DynamoDB, Redis
**Cloud & DevOps:** AWS (EC2, ECS, S3, Lambda, CloudWatch, ECR), Docker, Kubernetes, Terraform, Helm, GitHub Actions, CI/CD, Linux/Unix

## Work Experience

**Tata Consultancy Services (TCS)** Apr 2021 – Oct 2023
*Full Stack Systems Engineer | Client: GE* *Remote / Hybrid*

- Designed and developed Java **Spring Boot microservices and REST APIs** supporting customer onboarding, account management, and payment processing, handling over **2M+ financial transactions monthly**.
- Architected and deployed **Kafka-based event-driven microservices** for transaction processing, audit logging, and asynchronous notifications during monolith decomposition, reducing inter-service processing time by **40%**.
- Integrated external **AI/LLM services via REST APIs** with reusable provider abstractions, prompt orchestration, response validation, exponential retry policies, and automated failover for resilient multi-model integration.
- Engineered an enterprise **RAG knowledge retrieval service** using **FAISS, BERT embeddings, and LLM APIs**, boosting contextual search accuracy by **80%** across internal technical documentation.
- Optimized application performance via **MySQL query tuning, MongoDB indexing, Redis caching, and JVM tuning**, reducing API latency by **30%** and increasing throughput by **25%** under high concurrency.
- Built backend data migration pipelines using **Java, Python, and Apache Airflow**, migrating **50M+ records** with zero production downtime and verified data integrity.
- Containerized microservices with **Docker and Kubernetes**; automated deployment workflows via **Terraform, Helm, and GitHub Actions CI/CD**, cutting release cycles by **70%**.

**Launchpad Technology** May 2020 – Apr 2021
*Software Engineer* *Mumbai, India*

- Collaborated with cross-functional teams to engineer scalable backend APIs and full-stack features using **React, Node.js, MongoDB, and RESTful services** serving **270 educational institutions**.
- Implemented secure **JWT-based authentication**, asynchronous task processing, reusable service components, centralized exception handling, and structured logging, reducing code duplication by **35%**.
- Architected relational data models and schema migrations in **PostgreSQL/MySQL**, optimizing database indexing and connection pooling to reduce average API response latency by **25%**.
- Built automated **UNIX/Bash** shell scripts for server maintenance and continuous integration deployments across staging environments.

## Education

**California State University, Chico** | *M.S. in Computer Science* Chico, CA | May 2026
**University of Mumbai** | *B.E. in Computer Engineering* Mumbai, India | May 2020

## Technical Projects

**CoderAI – Open-Source Multi-Agent Code Intelligence CLI** *Python, MCP, Multi-Agent Systems, Docker*

- Engineered an autonomous multi-agent engineering assistant (*Planner, Coder, Reviewer*) for automated repository inspection, code synthesis, bug localization, and iterative repair.
- Integrated tool calling and secure sandboxed execution via the **Model Context Protocol (MCP)** to safely execute CLI commands, run test suites, and validate patches.

**TurboQuant: KV-Cache Quantization for LLM Inference** *PyTorch, CUDA, Quantization, Python*

- Implemented FP16→INT4 KV-cache quantization in **PyTorch and CUDA**, slashing inference memory consumption by $\sim$**4**$\times$ to support extended context windows on memory-constrained GPUs.
- Benchmarked end-to-end throughput, token latency, VRAM allocation, and perplexity across asymmetric/symmetric quantization strategies under simulated long-context workloads.

**AI Context Router** *FastAPI, Python, Docker, LLMs, Multi-Model Routing*

- Built an intelligent routing engine that dynamically dispatches user prompts to optimal LLM providers based on reasoning complexity, target latency, token overhead, and inference cost.
- Developed high-throughput asynchronous services using **FastAPI and Docker** with health monitoring and active error recovery.